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COMPARATIVE ANALYSIS OF ARMORED CONFLICT EXPERIENCE

A Draft Final Report

submitted by the

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I. METHODOLOGY

Selection of Armor Actions for Study

The criteria for actions to be studied were those set by Systems Analysis, Department of Defense (SA/DOD), to meet its research needs. They were as follows:

1. The actions were to be primarily tank-against-tank. The client expressed strong interest, however, in the effect of supporting arms upon such armor battles.

2. If possible, large numbers of tanks should be involved, preferably an armored division or more on each side.

3. Terrain must be relatively open; no actions in jungle or excessively mountainous terrain should be considered.

About halfway through the study, in further discussions with representatives of SA/DOD, the following additional desirable criterion was set forth:

4. If possible, at least one action treated should be a battle in which one side was defending prepared positions against a determined major attack.*

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*SA/DOD also expressed interest in wearout and mechanical breakdown during or at the end of periods of breakout and pursuit. The study participants agreed, however, that since such investigation required a very different kind of research process from the detailed investigation of individual armor actions already approaching completion, it was not feasible in the time available to attempt it. Such data could best be obtained, it was agreed, by following one or more armored divisions day by day through a long period including periods of both combat and noncombat.

As a basis for selecting the actions to be studied, an overall list was first prepared of a large number of varied armor actions in World War II.* The list was drawn up by the senior study participants in a planning conference for the study. It included five Northwest Europe actions during 1944-1945, three Mediterranean Theater actions, three involving British tanks in the North African Western Desert, six Eastern Front actions, and three other operations--the German invasion of France and Belgian Flanders in 1940, the India-Pakistan tank engagements of 1965, and the Israeli invasion of the Sinai Peninsula in 1956.

It was decided that the actions to be treated in the study would be chosen from this overall list on the basis of availability of records to provide the detailed strength and loss data needed for meaningful quantitative analysis. The results of record searching at the Federal Records Center, Alexandria, and at the US Army Armor Center at Fort Knox, together with consultations with historians of the Office, Chief of Military History, and the Document Reference Library of the Research Analysis Corporation, resulted in the following decisions.

1. Eastern Front. The tank battles on the Eastern Front would have been ideal for purposes of the study because of the large numbers involved and the open terrain. Unfortunately, it

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*Korea was excluded because of the small numbers of tanks involved in individual actions.

was conclusively determined that there are no German records available in this country that deal with events at the battalion level, and no detailed strength and loss figures even at division level.* Soviet records, of course, are not available. Late in the study, a study participant did prepare a paper on the Eastern Front battle of Targul Frumos, Rumania, on the basis of B.H. Liddell-Hart's interview with General Hasso von Manteuffel, who commanded the German forces in the battle. Although the account was one-sided, and probably not fully objective, it appeared to be sufficiently accurate and specific to be included.

2. Northwest Europe.

a. Operation GOODWOOD, August 1944, was chosen because of its close conformity to the criteria for the study, despite the fact that paucity of records would make thorough quantitative analysis impossible. The operation involved large numbers of tanks (more than 900 British tanks), in a determined attempt to break through a well-prepared defense in depth. There are no British primary sources (after action reports, etc.) in this country, and no German records bearing on the action in this country. British studies of the action that are available are focused in such a way that they do not provide the data needed for this study. Adequate

*Such records apparently were in this country, but were returned to West Germany without having been microfilmed.

records to complete this substudy are available in England and in West Germany.

b. The action near Arracourt, September 1944, was selected because of relatively good official records and availability of participants for interviews.

c. The US defensive actions in the "Battle of the Bulge" at St. Vith and Bastogne were chosen as examples of armor against armor in defense. More documentary data was available for St. Vith than for Bastogne, but the drama of both actions aroused so much interest at the time that a good deal of material based on survivors' accounts has been published and was utilized.

d. Good material was found of the Saar Campaign of November-December 1944, and this action, not on the original list, was added.

e. No data could be found on the German attack in Alsace, January 1945, and it was removed from this list. It was decided that it was not feasible to treat the breakout-and-pursuit period (Avranches-Falaise) of August 1944, as too many confused, individual engagements were involved to make analysis possible. As noted above, it was decided that study of breakout and pursuit, and the presumably attendant problems of mechanical reliability and endurance, could be better treated in a different kind of study.

3. Mediterranean. A well-documented small-unit action at Chouigui Pass in Tunisia, November 1942, was selected. Data on the other Mediterranean actions that had been considered was not adequate, due to nonavailability of German records.

4. Western Desert. British defense of Alam Halfa against Rommel in summary 1942 was chosen. It was felt that since it was a determined tank defense in prepared positions against a vigorous assault it was especially well suited for the study.

5. Other operations. Through the cooperation and help of the Defense Intelligence Agency, reasonably good data was available for the India-Pakistan actions, and a classified paper was prepared. The France-Flanders campaign, which was treated largely on the basis of published secondary materials, was found to provide considerable material for analysis, with adequate cumulative figures available for the whole campaign, and enough data available for specific operations, down to the small-unit level in one case, to provide accounts of the campaign at four different levels.

The ten actions finally selected, regrouped on a geographic rather than theater-of-operations basis were as follows:

A. Northwest Europe, World War II

1. France-Flanders, 1940
2. GOODWOOD, 18-21 July 1944
3. Arracourt, 19-22 September 1944
4. The Saar Campaign, 8 November-7 December 1944
5. St. Vith, 17-23 December 1944
6. Bastogne, 18-20 December 1944

B. North Africa, World War II

1. Alam Halfa, 31 August-3 September 1942
2. Chouigui Pass, November 1942

C. Eastern Front

1. Targul Frumos, Rumania, May 1944

D. Post-World War II

1. India-Pakistan, August-September 1965

Assembling the Material

The data from US records on the tank actions selected was collected by those senior study participants who had been given responsibility for preparing papers on the actions (Andrews, Crosby, Nihart) with the help of research assistants. The research assistants, with the help of the Federal Records Center staff, searched the records available at the Center for the units involved, for the time periods of the actions. After action reports, division histories, unit diaries, journals, and periodic reports (G-2, G-3, G-4) were the basic sources for strength and loss figures. The contemporary accounts of the actions were also useful, as were comments on the effectiveness of US and enemy tanks.

German records in this country are discussed in Appendix C, attached. The German records were searched by a German-fluent senior staff member of HERO, assisted by a German-fluent research assistant.

A trip to the Fort Knox Armor Center by one of the senior participants resulted in the selection and loan to HERO by the Armor Center of 21 unpublished monographs, several published works that had been difficult to obtain in this area, and 14 after action reports, other unit reports, and miscellaneous papers. This material was all examined, and much of it was found useful for the study.

The staff of the Historical Services Division and of the Histories Division of the Office, Chief of Military History, made available a number of unpublished monographs that were especially helpful in filling in the German side of several actions. Especially valuable was the advice of these experienced historians and researchers as to which materials were available elsewhere, and which simply were nonexistent or not available in this country.

The staff of the Document Reference Library of the Research Analysis Corporation made available, and in several cases borrowed for the study participants, various materials, especially classified studies on the use of armor. These, while generally adding nothing to the data for specific operations, were valuable for general background, including as they did such material as overall statistics on tank losses by causative agent, theoretical work on tank roles, and efforts to derive desirable strength ratios of opposing tanks. Other similar material was borrowed for the study

participants by SA/DOD. Much of this material is listed and briefly discussed in Appendix B (contained in classified annex).

About a dozen personal interviews with action participants were also conducted by one of the study participants (Colonel Crosby). Although these could not, of course, yield numerical data on strengths and losses, they brought out several otherwise unrecorded and very significant factors in the battles (as, for instance, the use of white phosphorus at Arracourt). Action participants were able to suggest sources of data in some cases, and at least one action participant had personal records in his possession that did give some detailed numerical data.

Published secondary sources were also used, the most useful being the Office, Chief of Military History, volumes in the series "US Army in World War II."

A good deal of research was also required to obtain in detail the characteristics of the tanks that took part in the actions studied. A most valuable source for German World War II tanks was a mimeographed British work, with photographs, "Illustrated Record of German Army Equipment, 1939-1945," Vol. III (M.I.10, The War Office, 1947) lent to HERO by RAC. B.H. Liddell-Hart's book, The Tanks, contains less detailed charts of characteristics for World War II tanks of all nations except the Soviet Union. Armor: A History of Mechanized Forces by R.M. Ogorkiewicz (New York: Praeger, 1960) provided some useful information on Soviet tanks,

but no detailed data. Handout data sheets published by US Army Ordnance on US and foreign tanks were also used. Contemporary US intelligence reports on German tanks, found at the Federal Records Center gave tank characteristics as observed in combat and derived from captured tanks. In general, the sources agreed closely. A challenging research problem was the determination of which model of the eight models of the German Pz.Kpfw. IV, for example, was involved in a given action in which the sources report only "Mark IVs." Data on development and dates of introduction of the various models, given in "Illustrated Record of German Army Equipment" and Liddell-Hart, together with slim clues in contemporary accounts of the actions, and other sources, were helpful in satisfactorily resolving this problem.

Interpretation and Use of Data

Each of the ten actions treated has been presented in a paper that includes the following parts:*

1. A narrative account of the action.

2. An analysis of the action, with emphasis on the factors determining success of the winning side.

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*Not all these items have yet been supplied for all actions, but all will be in the final report.

3. A map of the battle area.

4. Numerical data on tank strengths and losses, presented in tabular format.

5. Data on tank characteristics for all types of tanks involved in the action, presented in tabular format.

6. A bibliography of sources.

The numerical data of items 4 and 5, above, have been presented in tabular formats adapted from samples provided by SA/DOD. One format shows tank strength and loss figures for a given action, and the other shows characteristics of the tanks engaged in the action. Following the suggestion of SA/DOD, the actions have been divided into phases in cases in which they fell naturally into such phases, and in which there was data in adequate detail to make such phasing reasonably accurate. (None of the phased charts is yet ready for submission, but they are being prepared for the final report. Preparing tables for each phase of an action raises some difficult problems, such as determining the frontage of an engagement phase.) It was also decided to present the characteristics of all tanks involved in a given engagement in one spreadsheet chart with the account of that engagement.

In most cases there were gaps in the data that could not be filled except arbitrarily. For example, it was known in one action that both heavy and medium tanks were engaged, but it was not possible to tell how many of the tank casualties were heavy and

how many were medium. In no case except that of GOODWOOD did lack of data make it impossible to gain a clear picture of relative strengths and losses and to derive effectiveness quotients by use of the formula provided by SA/DOD. In the case of GOODWOOD, it was decided to submit the action without the unavailable German figures because of its special interest for the study.

In interpretation of data, a consistent attempt has been made to focus on two key points, central to the study and to the client's interest: (1) the advantage gained in each action from numerical superiority; (2) the advantage gained from qualitative superiority in specific tank characteristics. In a number of cases it was found that superior tactics, leadership, training, or good fortune was more significant in the outcome of the battle than either numerical superiority or overall qualitative superiority. In these cases, the question asked was, "What factors of numerical superiority, or what specific tank characteristics made it possible to successfully implement the superior tactics, or take advantage of the superior leadership, training, or luck?"

II. SUMMARIES OF TANK ACTIONS INVESTIGATED

France-Flanders, May-June 1940

The Germans invaded the low countries and France in May 1940 and quickly won a stunning victory, despite substantial numerical inferiority (3,609 French and British tanks to 2,574 German tanks). The German tanks were also generally inferior in armor and firepower, although they were somewhat superior in mobility--and mobility in all its aspects was the key factor in this campaign.

The well-conceived German strategic plan called for a thrust down through the low countries which would draw the best French and British forces, whereupon the main German attack, 44 divisions strong and using most of the available German armor, would push through the Ardennes area and break through to the Channel, cutting off the Allied forces to the north. The plan worked smoothly. The French had thought the Ardennes impassable for armor, and had defended it so lightly that its roads were entirely adequate for the German tanks. The Meuse crossings were forced and the breakthrough accomplished with the support of strafing Stuka aircraft, whose devastating psychological effect substituted for the artillery which the German armored units had outrun.

German Breakthrough and Exploitation North of the Somme, 10-24 May 1940

In this classic campaign the Germans concentrated 2,683 tanks, which qualitatively were somewhat inferior in gunpower and

and armor although superior in speed and range, against 2,160 French and British tanks. German losses were 1,080, an estimated two-thirds of which were repairable, to Allied losses of 2,060 which in addition to combat losses, included total losses due to capture or destruction to avoid capture. Determining factors were the German exploitation of concentration at the decisive point; tactical, operational, and technical surprise; and mobility in all of its forms, physical, mental, organizational, doctrinal, command-control-communications, and the use of firepower and terrain. Conversely, the Allies, although possessing an overall greater number of tanks in France (3,609 Allied to 2,574 German), dispersed their tanks in France, in the area of the campaign, and organizationally. Allied tactics, doctrine, command-control-communications, and mental preparation was also deficient for this new form of warfare. This 2-to-1 disparity of tank losses, while in favor of Germany and significant, does not tell the complete story, as this campaign destroyed France as a military power and drove Britain from the continent.

Rommel's 7th Panzer Division,
10-24 May 1940

The participation of Rommel's 7th Panzer Division in the campaign north of the Somme River provides a closer look at armored combat and its result. Rommel collided with both a French armored division and a British tank brigade, paced the

overall German attack (along with Guderian), and had a better tank loss ratio than the aggregate of all German panzer divisions. This study estimates Rommel's losses for this campaign at 72, while losses inflicted by him on the Allies were 161. Indices of effectiveness were thus 2.47 for Rommel and .4 for the Allied units encountered. The same determinants obtained in this portion of the campaign as for the campaign as a whole.

A Group Franc South of the Somme, June 1940

This small-unit action had a well-led and integrated French combat team launching a mobile counterattack from a defensive base and inflicting disproportionate losses on a German unit that had three times as many tanks but lacked reconnaissance or support. The French Group Franc consisted of 5 Somua tanks heavily armored for the time and armed with a very powerful 47mm. gun, 5 Panhard armored cars armed with 25mm. guns, 2 47mm. antitank guns, 2 25mm. antitank guns, heavy machineguns, and a motorcycle reconnaissance company. The Germans had 14 Pz.Kpfw. IV tanks with 30mm. of armor and a short 75mm. gun. The German tanks were the most powerful they possessed at the time, yet the French 47mm. guns easily penetrated them, while the German 75mm. gun of that time was only rarely effective against the 40mm. armor of the Somua. The Germans, without support or prior reconnaissance, blundered into the well-prepared and coordinated French defense and in fighting their way

out of contact and around the position lost 4 tanks to the 47mm. antitank guns and 3 tanks to the Somua tanks' 47mm. guns while knocking out one Somua--a score of 7 to 1 in favor of an integrated organization employing appropriate tactics. French effectiveness was thus 20, while German was only .05.

GOODWOOD, 18-22 July 1944

Operation GOODWOOD was a part of the Allied effort to break out of the Normandy beachhead in the vicinity of Caen. It is remarkable both for the effectiveness of preliminary aerial bombardment and for the misuse of tanks. British losses were very heavy (459 tanks lost or 36% of all engaged). German losses for this operation have not been found in the microfilmed records in the United States. The battle began with an extensive aerial bombardment which limited the initial response of the Germans to the British advance. However, as the British pushed further into the German defensive lines several weaknesses in their operational planning became apparent, while German defensive tactics were excellent. German armor and antitank guns were well placed to hit the advancing British tanks, which were trying to force the breakthrough without the aid of infantry. The British armor advanced beyond the range of supporting artillery fire; and the narrow attack zone bounded by minefields limited the number of tanks that could be committed and caused a severe traffic jam. GOODWOOD drew German armor off the US front and destroyed some of it; no breakthrough was achieved, but a continuation was made to the success of the later US breakthrough. British tank superiority (at least 4-1) and effective air bombardment resulted in a limited British advance.

Arracourt, 19-22 September 1944

This tank battle near Arracourt, France, was won by Combat Command A of the US 4th Armored Division through superior leadership and tactics, despite a German numerical superiority of 5:3 (in the medium tanks that actually fought the battle), and despite the overall qualitative superiority of the German Mark IV (Pz.Kpfw. IV) and Mark V (Pz.Kpfw. V) tanks to the US M4. Responding to a German counterattack by the 111th and 113th Panzer Brigades, the US forces achieved a US-German loss ratio of 4:1 (87 German tanks lost to 21 US tanks lost). US leadership effectively used the principles of mass and maneuver, concentrating ~~their~~ tanks and launching repeated spoiling attacks, while the Germans wasted their tanks in piecemeal commitments.

The superior mobility of the US M4 on dry, firm ground, and its power-traverse, quickly turning turret, more than compensated for the greater penetration capability of the German tank armament --75mm. guns, 43, 48, and 70 calibers long. Low visibility due to fog prevented US air support that would otherwise have been available, but the fog also allowed the Americans to close to a distance at which their less powerful L/40 guns became effective. Well-coordinated use of the highly effective tank destroyers and of artillery support contributed to the success. US use of white phosphorus as a first round allowed quick adjustment of fire, forced

several German crews to abandon their tanks, and allowed the US tanks to draw close without being seen.

The Saar Campaign, 8 November-7 December 1944

Bad weather was the most influential factor in the Saar Campaign, November-December 1944. Continual rain made the ground nearly impassable and poor visibility prevented air support. The aim of the US Third Army offensive was to clear the southern part of the Saar Valley and eventually to establish a bridgehead over the Rhine. In this operation, the US 4th Armored Division (271 tanks) was opposed by 9 German divisions, 2 brigades, and some miscellaneous units (145 tanks). The German Mark IVs, Mark Vs, and Mark VIs were superior to the US M4s in flotation, gunpower, and armor protectin. Despite their smaller force, the Germans fought a successful delaying action. They suited their tactics to take full advantage of the geographic and climatic conditions, while --because of these circumstances--the United States was engaged in an offensive for which tanks with marginal flotation were not well suited. Including 10 tanks that were recovered and returned to service, the US losses were 56 tanks to the German loss of 61. Thus although the Germans were outnumbered in tank strength almost 2:1, tank losses were almost equal for the two sides. Ten US M5s and 45 miscellaneous German tanks that were present were not suitable for action against opposing armor and antitank weapons, and if they

are eliminated from consideration the German losses are actually less, by a ratio of 1-4, than those of the US force.

St. Vith, 17-23 December 1944

The mission of CCB of the 7th Armored Division at St. Vith was to hold at all costs in order to disrupt the German Ardennes offensive. Through use of excellent defensive tactics it did hold for six days against greatly superior German forces, delaying the Germans long enough to make certain the failure of the German offensive. Inadequate German planning allowed the German offensive spearheads to be held up by traffic jam, so that they reached St. Vith after the 7th Armored. The Germans committed their tanks piecemeal for three days, wasting the advantage of their numbers. CCB used a mobile defense, defending dominant terrain with dug-in infantry and tank destroyers, while holding its tanks as a mobile counterattacking force. Repeated US spoiling attacks kept the Germans off balance and made them believe they faced a much larger force than they did. The mobility of the M4 tank permitted the successful implementation of these tactics. /Loss ratios to be included when charts are completed./

Bastogne, 17-20 December 1944

In the area east of Bastogne, in southeastern Belgium, on the edge of the Ardennes Forest, elements of two American armored divisions, the 9th and the 10th, clashed with elements of three

German divisions, the 2d Panzer, 26th Volksgrenadier, and 130th Panzer Lehr in a series of uncoordinated actions. The German XLVII Panzer Corps, to which the German divisions belonged, was driving westward around Bastogne toward the Meuse River, as part of their great Ardennes counteroffensive. The Americans were assigned the task of holding them back from the roads leading into the city. CCR of the 9th Armored Division and CCB of the 10th were posted at various points in task forces comprising approximately a battalion apiece, variously supported by artillery, armored infantry, and other elements. Most of the action took place in fog, smoke, or darkness, with the Germans attacking with small numbers of tanks, and generally at fairly close range since poor visibility prevented their taking advantage of their longer range guns. The greatly outnumbered Americans suffered losses in a ratio of 5 to 3, but they achieved their objective of delaying the German advance. Tanks, tank destroyers, artillery, and infantry worked together in teams to drive off repeated attacks or, to exact a high price for their own destruction. The Germans, surprised to find their way blocked, lacked the information and the leadership to coordinate their attacks and to remove quickly the obstacles which they certainly had the strength to overcome. They did ultimately get through, but not until their schedule had been disrupted and their attack thrown off balance.

Alam Halfa, 31 August-3 September 1942

Elements of the British 7th and 10th Armored Divisions conducted a successful defense of Alam Halfa in the Egyptian Western Desert against General Erwin Rommel's Afrika Corps (15th and 21st Panzer Divisions) by using their tanks dug in, in a static defensive role. This unconventional use of armor was dictated by the qualitative inferiority of the Grant medium tanks with which the British were equipped. The Grants had 75mm. short guns (31 cal. long) with relatively low muzzle velocity, and speed not significantly higher than that of the opposing Pz.Kpfw. IIIs and IVs, which were much more heavily gunned (50mm., L/60 and 75mm., L/43 or L/48). The usefulness of the Grant gun was further severely limited by its low side mount, which permitted little adjustment of elevation or traverse. The Grants' high silhouettes also made them easy targets on the battlefield and, as the British used them, the tank commander had to partly expose himself in order to direct fire.

A breakthrough at Alam Halfa was essential to Rommel, as he could no longer support himself in Egypt unless he could break through to Alexandria and Cairo. Thus, although the numerically superior British force lost tanks in a ratio of 1.34:1, Rommel's repulse was not only a British tactical success, but also a significant strategic victory.

An important factor in the battle was a British minefield that delayed the Germans for six hours, forcing Rommel to attack in daylight and lose the advantage of surprise, and also so depleting his limited fuel supplies that he was forced to attack frontally, rather than make the flanking sweep he had planned. British air harassment also contributed significantly to the victory.

Chouigui Pass--26 November 1942

Tank speed and maneuverability, together with a near-perfect tactical plan, were the main factors influencing the small-unit US victory at Chouigui Pass in Tunisia, the first US armor action of World War II. In overall qualitative characteristics, the German tanks outclassed the American tanks. A detachment of the German 190th Panzer Battalion, 10th Panzer Division, was fighting with 13 Mark III and Mark IV medium tanks, all of which were equipped with 50mm. and 75mm. guns. The US force--two companies of the 1st Battalion, 1st Armored Regiment, had 25 tanks, all light M3s, with 37mm. guns, 53 calibers long that were completely incapable of penetrating the German armor. The mobility of the light tanks, coupled with superior tactics, enabled the Americans to hit the German tanks in the rear, engines, and tracks, resulting in a US loss of only 28% of the tanks engaged, and a German loss of 69%. The US to German loss ratio was 1:1.29.

Targul Frumos, May 1944

A German Panzer Division under General Hasso von Manteuffel and a Soviet tank army (or motorized army) opposed each other on a broad open plain in the Battle of Targul Frumos. Speed and maneuverability were the important factors and in these the German Mark IVs and Mark VIs surpassed the powerfully gunned Soviet JS-III. The Russians were able to surprise the Germans by opening fire with their high-velocity 120mm. gun at a range which was far beyond the effective range of the German tank guns. When the Germans were within their normal firing range their guns were unable to penetrate the heavy Soviet armor. The Germans closed the distance quickly and, due to superior mobility, maneuvered around the Russian tanks to hit their thinner side and rear armor. The result was a clear-cut German victory, achieved by superior leadership and tactical ability and facilitated by certain weaknesses in the JS-III--a slow hand traverse mechanism for a very heavy turret, a meager supply of ammunition because the rounds were so large that fewer rounds filled up more space, and heavy separate hand-loading ammunition that slowed the rate of fire. Of a total of 410 tanks the Russians lost 350 and 60 others were damaged; the Germans lost only 11 tanks out of a total of 160.

III. ANALYSIS

A study of the armored force engagements contained in this study suggest that there are three major kinds of tank operational characteristics, and two important subsidiary characteristics, which affect the outcome of armored engagements. These five characteristics, listed in what appears to be the order of their relative importance, are as follows:

1. Tactical Mobility (Substantial speed combined with cross-country capability under any likely combat circumstances, to include if possible a slight margin of superiority over enemy tanks of comparable role and characteristics.)

2. Firepower (Ability to fire quickly, accurately, and repeatedly, with a weapon whose projectile has sufficient mass and muzzle velocity to destroy a likely opposing tank at reasonable range.)

3. Protection (Full protection against small arms, shell fragments, and long-range fire from the most likely possible armored opponent.)

4. Endurance (Ability to operate effectively over substantial periods of time and extensive distances; this is also an aspect of strategic mobility, and is enhanced by well-organized logistical support.)

5. Reliability (Minimum of breakdowns under the combat circumstances to be expected; ease of first echelon maintenance.)

Also emerging from the experiences in this study there appear to be five major prerequisites for the effective employment of armored forces:

1. Aggressive, imaginative, skilled leadership, to exploit one's own advantages and minimize one's own weaknesses with respect to the opponent.

2. A coherent and effective doctrine for the employment of armor.

3. Effective training in the established doctrine to assure common understanding and maximum coordination capability.

4. Maximum possible combined arms coordination with available artillery, infantry, and air support, in accordance with the established doctrine.

5. Employment of tanks in mass, to the maximum extent possible under the circumstances.

From these experiences it is possible to make several general observations:

1. Numerically fewer tanks, with superior operational characteristics, are particularly effective in the defense, although this effectiveness will be seriously reduced if they do not have firepower superiority over the enemy.

2. Numbers, in general, are more important than firepower or protection, because of the advantages superior numbers provide in maneuver and mass. Number, however, are evidently less important than mobility, unless the numerical inferiority is overwhelming.

3. In engagements of large numbers of adequate tanks, the side with the greatest mobility is usually victorious, unless it is greatly outnumbered, or unless its tanks are seriously inferior in one of the three major operational characteristics. The turning point in terms of the relationship of mobility to numbers depends on too many factors to be specified precisely, but the effects of disparity of numbers are obviously reduced by effective doctrine, adequate training, and good leadership on the side with numerical inferiority.

4. The effects of numerical inferiority of tanks are greatly minimized by the availability of supporting forces of other arms, particularly antitank weapons, artillery, infantry, and air support. These supporting weapons and units have the primary effect of enhancing the mobility of friendly tanks, and inhibiting the mobility of the opposition.

Not surprisingly, this study shows that, all other things being equal:

a. Superior leadership is usually successful.

b. Superior numbers usually win.

c. Superior weapons usually defeat inferior.

The study also suggests certain conclusions with respect to various mixes of leadership, numbers, and weapons. If all the variables and intangible factors of combat (leadership, terrain, morale, availability of supporting troops, etc.) are equal, and if the size of the forces is approximately equal, tanks with superior characteristics will ordinarily defeat an equal number of tanks with lesser characteristics. If the tanks of one side have a superiority in one or more characteristics, while those of the other side have superiority in other characteristics, the outcome of a conflict between equal numbers is much harder to predict. The results of engagements studied would suggest, however, that if the tanks on both sides meet at least minimum operational requirements for adequacy in all five operational characteristics, then the listed order of priority of the five characteristics (mobility, firepower, protection, endurance, and reliability) become particularly significant. There should be exploration of the possibility of assigning relative quantified values to these characteristics, for comparative purposes.

It is understood that the purpose of this study is to assist policymakers in deciding an allocation of available resources to obtain the maximum return for armored conflict. Thus, it is essential at the outset to determine how and why tanks can contribute to success in battle. There see to be three major possible roles for tanks, not mutually exclusive, and not necessarily

in order of relative importance: (1) to defeat other tanks; (2) to provide the most economical collection of protected firepower for defense; (3) to contribute to maximum overall combat capability in the maximum possible variety of combat situations.

It is undeniable that the best antitank weapon is a tank--but it is an extremely expensive weapon. The resources of men, complicated manufactured materials, and money required for production, training, and deployment could produce a number of high-velocity antitank guns, tank mines, and the highly mobile troops required to man these guns and place these mines, which would in sum be more effective in antitank defense than would the single tank they replace. Thus, if our mission is simply to defeat other tanks, the best allocation of resources might well be to antitank means other than tanks.

It is clear, however, that an enemy tank force meeting such a defense of antitank guns and mines would employ a combined arms approach in which infantry, artillery, and engineers, supported by tanks, would be employed.

To counter such a combined arms offensive against our antitank defenses, it would be essential to have, in addition to the purely antitank defense, adequate quantities of defending infantry and artillery, plus, probably, a small number of very powerful tanks, particularly strong in firepower and protection to work with infantry and artillery in opposing hostile attacks and to

provide an effective counterattack force against the enemy's combined arms capability. There is reason to believe, however, from the examples we have studied, that if the attacking force can at least offset the capability of the defense in artillery, infantry, and engineers, and can mass a superior number of tanks with adequate operational characteristics (even though lacking all or most of the qualitative values of the less numerous defending tanks), this qualitative superiority of the defenders is not likely to be sufficient to assure successful defense. At the same time, even if the defense should be completely successful, there is some question whether the smaller number of very powerful tanks would be able to exploit such a defensive success as well as a larger number of somewhat less expensive tanks.

While tanks are highly effective as antitank weapons, it is clear from our studies that they are also especially effective, both as weapons and as components of a combined-arms team, in employing their mobility and shock action capability in offensive action against both unarmored and armored hostile forces. A successful defense is at least partially wasted if the defenders are not thereupon able to seize the initiative in offensive action. It is clear from the examples we have studied that a smaller number of very powerful tanks, particularly if not more mobile than the opponent's, is not likely to be so successful in an attack against enemy armor as would be a larger number. Furthermore, it

appears to have been amply demonstrated that the shock action and exploitation capabilities of tanks against unarmored forces are enhanced by numbers, probably in geometric proportion, in terms both of psychological impact and of maneuver capabilities.

It is probably impossible to come up with a formula which will express the ideal relationship between numbers of tanks and quantities of unarmored units armed with small arms as well as with heavier weapons, including artillery. In the type of relatively open terrain which favors any use of armored forces in large numbers, however, it is probable that a proportion of one armored division to each other type of division would not be too much armor. Since it is doubtful that any nation will have the resources to cover vast fronts with this high proportion of armor to infantry and other arms, the ratio could be considered as an optimum objective. It would thus appear that resources available for armor along should be so allocated as to produce maximum numbers of tanks that are adequate in the four lower priority characteristics, and that are highly mobile under all likely combat circumstances. It would appear to be a serious mistake to try to approach the optimum in all five characteristics, if this were to mean a substantially smaller number of tanks.

An important factor in the decision must be the realization that the role of armored forces cannot and should not be limited to combat with opposing armor. We have noted that nonarmored

antitank defenses can be made far stronger than would be possible by allocating the same resources to armor for defensive purposes. More significant, however, is the demonstrated fact of two more important, and overriding roles for armor: (1) extensive, rapid, and relatively secure maneuver capability in intensive combat; (2) the psychological effect of the employment of tanks in mass against hostile unarmored forces; and (3) breakthrough and exploitation in rear areas with relative imperviousness to rear areas troops and countermeasures. Thus in allocating resources the aim should be a combination of highly mobile, but otherwise merely adequate, armor in combination with effective antitank weapons, artillery, and infantry, to be employed both offensively and defensively (bearing in mind that firepower is the most important operational characteristic of armor in combined arms defense). But it should be further borne in mind that the primary role of the tank is offensive, rather than defensive, and that other, nonarmored measures are available to offset tank deficiencies in numbers and quality in the defense.

On balance, then, the effectiveness of tanks in their primary missions of maneuver and shock action is at least directly proportional to the number of "adequate" tanks available. This would seem to justify a production policy which will assure the maximum number of tanks which at least meet all five of the minimum operational requirements, and which exceed the minimum mobility requirement.

IV. EVALUATION OF ANALYTICAL FINDINGS

Limitations on Data and on Analysis

For reasons discussed earlier in this report (Section I, Methodology), it has been necessary to proceed as rapidly as possible to the analysis of as many individual tank engagements for which sufficient data was available to permit a reasonably valid analysis of armored conflict. While every possible effort was made to find actions representative of as wide a variety of tank engagements as possible, the initial approach was necessarily in the nature of a net-casting process, rather than a deliberate approach to analysis of the results of specific actions, or series of actions, as part of a planned and systematic procedure. Furthermore, lack of available data for some actions or campaigns which had been on the original list has led still further toward a collection of individual studies selected for convenience, rather than as elements of a sound, systems approach.

The engagements studied are for the most part those for which primary and secondary source material is readily available because the actions were in themselves dramatic or unusual, or were parts of larger, dramatic operations. Not unexpectedly, these are more often than not actions in which a high order of leadership, or special circumstances, resulted in victory by tank

units that were outnumbered, or were outgunned, or that otherwise suffered under some serious disadvantage. Fortunately, there are enough of these actions, in different armies, to provide some sort of balance, but this balance is largely fortuitous. It is not clear from our data whether the actions we describe were typical, or quite atypical, or something in between. We suspect the latter, but we cannot be certain from the data that we have.

This does not in any way invalidate the data, analysis, or the findings, of any of the individual actions investigated in this study, nor--we believe--does it invalidate the overall conclusions we shall draw from review of these findings in the next section of this report. But it does mean that we cannot be certain that we have all of the conclusions which might be desirable for the purposes of this study.

What is particularly lacking from this study is an indication of the effect of sustained operations on armored conflict, and on different kinds of tanks with varying characteristics, particularly in the kind of war of rapid and long-range movement for which armor is particularly suited. In the time available for this study there was not sufficient time nor funds to undertake the painstaking collection of basic primary source data on the operations of one or more units over extended periods of time, to include all of the opposing force data essential to the analysis.

That such data can be found seems certain, but even that for US forces is not readily available in this country, and in some instances it would be necessary to obtain access to German (and possibly British) archival materials.

As a result, we have not been able to give sufficient consideration to the operational characteristics of endurance and reliability, nor to the effects of sustained operations on leadership and on troops in their performance in armored conflict.

Systematic Approach to Remove Limitations

Four specific actions could be taken to develop a research plan which would both profit from what has been accomplished in this study, and would at the same time assure the comprehensiveness and catholcity which was not possible in this study in the time available:

1. Review each of the engagements examined in detail in this study to ascertain which ones could profit from expansion, and the availability of more data, and which ones could be related either to subsequent or to prior operations, in order to provide greater insights into the operational tank characteristics of endurance and reliability.

2. Ascertain what information is needed, and where it is likely to be available, for important armored conflict and operations experience in modern history which has not been covered

significantly in this study; for instance: primary source data is undoubtedly available on operations in the Western Desert from 1941-1943 both in Britain and in Germany; more information on armored combat on the Eastern Front is probably available in Germany.

3. Review the various armored conflict experience analyses available (See Appendix B) to find actions or campaigns in which data was evidently available to the analysts who prepared these studies, and select all instances which offer the possibility of fruitful further investigation along the lines of this study.

4. Select two or more armored units with extensive experience in World War II, and follow these for one or more entire campaigns, detailing and analyzing all noncombat as well as combat losses, and evaluating the influence of these losses--and all other relevant influences--on the results of engagements in which these units were engaged. Ideally, at least six such unit operational surveys should be undertaken: two American armored divisions, two British armored divisions, and two German armored divisions.

Implications for Cost-Effectiveness Military Operations Research

The laboratory of the military man has traditionally been war itself; from review and analysis of the records of past combat

It is evident, then, that further extensive research in all these records can perform two important kinds of functions. First, the records can be exploited systematically to provide the kinds of information needed to assist in important policy decisions on current military problems of procurement, design, or tactics, as in the case of this study. Second, these records can provide a basis for equally systematic verification or validation of the results of wargames.

There are a number of ways in which these records could be systematically mined and the results analyzed to accomplish these objectives. One possibility would be to establish an institute for military operational records evaluation, which might operate much as the Navy's George Washington University Logistics Research Project operates. This and other possibilities should be explored for the benefit of all of the military services, of the Department of Defense, and in particular the Office of the Assistant Secretary of Defense (Systems Analysis). Such an institute, furthermore, could provide bases for modified combat reporting systems, to remedy in the future reporting deficiencies of the past.

Appendix C

COMMENTS ON GERMAN MILITARY DOCUMENTS

(A Report by Wlodzimierz Onacewicz)

1. Examined microfilms of German military documents at the National Archives contain a tremendous amount of unpublished information on various aspects of the German military establishment during World War II, its organization and armament, strategy and tactics, and descriptions and analyses of battles and campaigns, evaluations of the enemy, and evidence of shortcomings and difficulties experienced by the Germans in the last years of the war. This information is contained in war diaries of divisions, army corps, armies, and army groups; combat and logistic orders and reports; intelligence documents; reports on human and material losses and replacements; analyses of battle experience; and a great variety of other documents. It must be observed that because of the unsystematic way in which the microfilmed files were compiled, the search for a definite kind of information requires examination of a considerable number of unrelated documents and is an extremely time-consuming process. However, we have learned that the results of such search are often rewarding.

2. Our recent specific search for information on German military losses, human and material, particularly for armored conflict, in most cases gave only limited results. This was due partially to limited time available for search, but mainly to the scarcity of reports on losses in the files. Our research proved beyond any doubt that in the German armed forces daily and periodical reports on human and material losses were systematically compiled by all German units and headquarters, and forwarded to the next level of command. When we found such reports (usually prepared on special printed forms), they proved excellent, detailed, complete, and apparently accurate, and were most useful to our studies. Trying to establish the cause of scarcity of losses reports, in the microfilmed files, we learned that a great number of German military files have been returned to Germany without being microfilmed, which suggests that among nonmicrofilmed and returned documents were reports on losses that, because they were so common, did not seem important to the persons who selected documents for microfilming. Thus it seems most probable that the missing reports on losses could be located in Germany.

3. Conclusion. Exploration of German military documents, as a primary source of information on the German side in World War II, is of great significance to our studies on the postwar development of various aspects of military doctrine, principles of modern combat, and combat effectiveness of modern weapons. In

fact, it seems that no study using the experience of recent wars can be considered sufficiently comprehensive without the study of these unique documents. Although some of this material has been retained in the US National Archives, systematic research in German records requires searching the archives in Germany.